

# Majani Relay

From a relay farmer's leaf photo to the officer's next visit

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Live app: [git-lsd.github.io/majani-relay](https://git-lsd.github.io/majani-relay)

Code: [github.com/Git-lsd/majani-relay](https://github.com/Git-lsd/majani-relay)

## 01 Problem statement

Because of this tool, **the county extension officer or cooperative agronomist will send their next visits to the villages with the most leaf rust, by the week the relay farmers' plot checks reach the cooperative**, that they would otherwise do late, after paper reports arrive; we know because one Kenyan extension officer typically serves 1,500–3,000 farmers and paper-based reporting has caused “delayed information flows” (Ministry of Agriculture draft data policy, 2026), and coffee-specific extension has “collapsed” in places (Coffee Development and Marketing Strategy, 2024).

## 02 Where it sits in the user's day

- **Relay farmer, on a plot visit** (a farmer the co-op trains to visit members' plots): asks consent, takes 15 leaf photos, plays each answer in Swahili. Offline.
- **Extension officer, on meeting the relay farmer**: labels the “not sure” photos, makes the final call, updates the model.
- **Co-op office, planning visits**: ranks villages by rust, so the officer goes first where rust is highest.

**The gap**: one officer per 1,500–3,000 farmers; late reports.

## 03 What the AI does, and why AI

- **Turns photos into counts**: names each leaf (healthy, rust, leaf miner, brown eye spot, Phoma) or says “not sure”.
- **Knows when it does not know**: an unfamiliar-photo check and an “other problem” answer.
- **Learns from the local officer's labels**, on the phone.

**Why not SMS, a spreadsheet or a search?** SMS cannot read a photo, a spreadsheet cannot look at a leaf, and a search needs signal and gives general pictures. A paper form holds each person's guess; the AI applies one rule on every phone.

### GUARDRAILS BUILT IN THE APP

- “Not sure” when unsure, unfamiliar or another problem; those photos wait for the officer, out of the count.
- The officer makes the final call; the relay farmer can disagree; 1 in 10 answers are spot-checked.
- A fixed list of 24 answers; it never writes its own text; no pesticide names or doses.
- Consent first; no names, phone numbers or locations stored; offline, on the phone.

## 06 Next: a 90-day pilot (proposed)

One Kirinyaga co-op, 5 relay farmers, one officer, the co-op manager.

- **Before**: per-officer PINs replace the demo PIN; an offline visit timed on a low-cost Android phone.
- **Week 1**: a Swahili-speaking officer reviews all 24 answers.
- **Weeks 3–8**: the first 200 officer-labelled Kenyan photos are sealed as a test; below 70% correct, AI answers switch off.
- **Week 12**: continue, change or stop, decided with the co-op.

## 04 Results at a glance

**91%**

ANSWERED

**Healthy and rust field photos** never trained on: answers 91%, and 89% of those are right; the rest go to the officer.

RoCoLe, Ecuador; cross-validated

**93%**

TO THE OFFICER

**New country**: 93% go to the officer, not a guess (forced: right on only 43%). After 100 officer labels: answers 46%, 94% of those right.

Uganda farm photos; dataset labels stand in for the officer

**3.4 → 1.0**

FALSE ALARMS

**Village ranking**: false alarms per 40 villages, with the small-sample adjustment. Misses near the alert line: 1.6 → 3.0.

Simulated villages, at measured error rates

## 05 Tech stack

- **Static web app (PWA)**: plain HTML, CSS, JavaScript; no server, no account.
- **On-phone AI**: MobileNetV3 backbone (16.8 MB, ONNX) + small head (82 KB) + familiarity set (1798 KB) in onnxruntime-web.
- **On-phone refit** from officer labels; updates shared by WhatsApp or Bluetooth.
- **Offline** after a first load of about 36 MB; records in IndexedDB.
- **Village ranking**: Beta-binomial statistics, not AI.

## 07 Our take: localizing AI

Localizing AI is not translating an app. Our first model, trained on lab photos, was right on 84% of held-out lab photos but on only 2.5% of Ecuador field photos, while 93% confident on average. So we retrained it on field photos, taught it to say “not sure”, and let the local officer teach it on the phone. **To us, localizing AI means the people closest to the farm teach it, correct it, and stay in charge.**